

sparsity of the power system increase. Furthermore, the effects of telecommunication speed among computers are examined.

Discussers: Dr. M. S. Kurzny and Th. Van Cutsem

References

- [1] H. Mukai and I. N. Katz, "Parallel multi-area static state estimation of electric power systems using sparsity," *Electric Power Problems: The Mathematical Challenge*, ed. by A. M. Erisman, K. W. Neves and M. H. Dwarkanath, SIAM, Philadelphia, 1980.
- [2] K. Seidu, *Parallel State Estimation and Meter Placement in Electric Power Systems*, D.Sc. Thesis, Washington University, St. Louis, 1983.

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Network Observability: Identification of Observable Islands and Measurement Placement

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The static state estimation for on-line applications in power system control centers processes a set of often redundant measurements to obtain the best estimate of the state of the system. A power network is observable if the set of measurements makes the state estimation possible. The observability depends on the number of measurements and their geographic distribution. Even if the system is designed to be observable, temporary unobservability may occur due to network topology changes or failures in the telecommunication systems.

In this paper, first an algorithm is presented which:

- (i) tests the observability of the network and
- (ii) identifies the observable islands of the network when it is not observable.

When the network is not observable, one would like to know where additional measurements should be placed so that the network will become observable. In system operation when temporary unobservability occurs, sometimes it is possible to add some pseudo-measurements to make the system observable. In such cases, it is desirable that the lower quality pseudo-measurements should not contaminate the state estimation of the observable islands. In this paper an algorithm is also presented for measurement placement which

- (i) selects a minimal set of additional measurements to make the network observable
- (ii) guarantees the additional measurements would not contaminate the state estimation results.

The observability analysis is performed during the triangular factorization of the gain matrix G . For identification of observable islands, usually one solution of a linearized state estimation is required where the triangular factorization of the

gain matrix was already done in the observability test. The measurement placement algorithm also makes use of the triangular factorization of the gain matrix and the solution of a linearized state estimation problem. Thus the observability analysis imposes very little extra cost in computation. Indeed it is a fraction of the computation required for the state estimation algorithm itself. The method is also stable and robust. The proposed algorithms use essentially the same subroutines already available in the existing state estimation program. The new approach of this paper reduces the observability problem to a subproblem of a more general state estimation problem, which means simplification in coding, reduction of interfacing problems, in addition to computational savings.

The two algorithms are based on the results derived in [1]. After the presentation of each algorithm in the paper, an illustrative example is worked out in detail. The algorithms have been tested extensively on all systems that have been studied for observability in the published literature. Some test results on a 14-bus network and a 121-bus network are included in the paper.

Discusser: Walter L. Snyder, Jr.

References

- [1] A. Monticelli and F. F. Wu, "Network Observability: Theory," submitted for presentation at IEEE PES Summer Meeting, Seattle, WA.

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Network Observability: Theory

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A complete theory of network observability is presented. The major points are summarized below:

- A network is said to be observable if, whenever there is any nonzero flow in the network, at least one of the measurements should read nonzero. In other words, a network is observable when all measurements are zero implies that all flows are zero.
- A network is observable if and only if the state estimation can be solved with a unique solution.
- An observability test can be performed based on the triangular factorization of the gain matrix. It goes as follows. In carrying out triangular factorization of the gain matrix G whenever a zero pivot is encountered, we add a θ -pseudo-measurement for the corresponding node, the effect of that is that the 0 is replaced by a 1 in the triangular factor. The network is observable if only one zero pivot is encountered in the process and not observable if more than one zero pivots are encountered.
- By setting the θ -pseudo-measurements of the nodes of zero pivots to different values and all the other measure-

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